

SLURRY_FREE_ACID Modeling

Project advancement and current evidence

Internship project

July 2026

1. The project now covers three operational cases

1. Direct future level

$$X_{\leq t} \longrightarrow \hat{y}_{t+h}$$

Process measurements predict the future acid level directly.

Useful when process measurements are available; evaluated against persistence when y_t is observed.

2. Target-anchored forecast

$$X_{\leq t}, y_{\leq t} \longrightarrow \widehat{\Delta y}_{t,h}$$

The model predicts the change, then reconstructs $\hat{y}_{t+h} = y_t + \widehat{\Delta y}_{t,h}$.

Best forecasting case, but it requires a current free-acid measurement.

3. Target-free virtual sensor

$$X_{\leq t} \longrightarrow \hat{y}_t$$

Process measurements estimate the current acid level without any target column.

Designed for a new CSV containing only process variables.

Important distinction

These are different deployment contracts. The target-anchored model cannot be used when the uploaded file contains no SLURRY_FREE_ACID observations.

2. Preparation was designed as a controlled experiment

Data and targets

- ▶ 44,640 chronological one-minute records
- ▶ 21 process measurements
- ▶ Forecast horizons $h \in \{1, 5, 10, 15\}$ minutes
- ▶ $y_{t+h} = \text{SLURRY_FREE_ACID}_{t+h}$

Leakage controls

- ▶ Chronological train/validation/test splits
- ▶ Lag selection uses training targets only
- ▶ Imputers, scalers and filters fitted on training only
- ▶ Structural warm-up removed before splitting

Scientific purpose

Compare feature hypotheses and horizons fairly, rather than create one oversized feature matrix.

3. Four feature sets test four hypotheses

Set	Contents	Hypothesis
A — Raw	Current process measurements	The current process state may be sufficient.
B — Temporal	Raw values plus selected lags, rolling behavior, differences and trends	Process history may reveal delayed responses and recent dynamics.
C — Process-aware	Raw values plus balances, ratios, thermal gradients and stability indicators	Interpretable process relationships may improve prediction and generalization.
D — Full	Accepted temporal and process-aware features	Historical and process-relationship evidence may be complementary.

Examples include TOTAL_ACID_FLOW, ACID_FLOW_IMBALANCE and TOTAL_SLURRY_M3_FLOW. Ratios and proxies remain provisional when units or stream correspondence are unconfirmed.

4. A small model set covered complementary behaviors

Model	Quick definition	Why it was included
Ridge	Regularized linear regression	Stable, fast and interpretable baseline for many correlated features.
PLS	Latent-variable linear regression	Classical industrial baseline for strongly collinear process sensors.
Extra Trees	Average of randomized decision trees	Captures nonlinear thresholds and feature interactions.
HistGradient	Sequential histogram-based trees	Compact nonlinear boosting with efficient training.
XGBoost / LightGBM	Optimized boosted-tree ensembles	Strong tabular learners for complex nonlinear relationships.
KNN	Prediction from similar historical process states	Tests a local, just-in-time modeling hypothesis.
Weighted blend	Validation-selected average of two models	Combines complementary linear and nonlinear errors.

Search spaces were deliberately small. Selection used chronological validation; the untouched test period was evaluated once.

5. Direct level models did not beat persistence

Horizon	MAE	RMSE	vs. persistence
1 min	0.269	0.332	-67.9%
5 min	0.275	0.338	-35.1%
10 min	0.280	0.344	-22.6%
15 min	0.304	0.378	-27.5%

Validation selected Ridge with Feature Set D at every horizon. Negative improvement means the model is worse than carrying the current target forward.

Interpretation

- ▶ The model learned the operating level.
- ▶ Persistence already has a powerful anchor: the observed y_t .
- ▶ Direct level learning spent capacity reproducing that level instead of learning movement.

Decision

Reframe the anchored task around the future change, not the absolute level.

6. Predicting change made the anchored forecast useful

Horizon	MAE	RMSE	R^2	vs. persistence
1 min	0.102	0.151	0.889	+23.2%
5 min	0.113	0.169	0.862	+32.3%
10 min	0.126	0.182	0.839	+34.9%
15 min	0.176	0.237	0.726	+19.8%

Selected model: Ridge, hybrid full set, 188 inputs, $\alpha = 10$. Metrics are computed after reconstructing $\hat{y}_{t+h} = y_t + \widehat{\Delta y}_{t,h}$.

Why it improved

- ▶ y_t supplies the current level.
- ▶ The model focuses on the smaller future movement.
- ▶ Target history and process features both added value.
- ▶ Ten minutes gave the largest relative gain, but with higher absolute error than 1 or 5 minutes.

Constraint: this model requires current and past SLURRY_FREE_ACID values at prediction time.

7. Process information and target history were complementary

Ablation findings

- ▶ Accepted features improved over raw-only inputs by **10.3–24.2%**, depending on horizon.
- ▶ Adding target history to exogenous full features improved RMSE by **8.2–12.7%**.
- ▶ Adding process/full information over target history alone improved RMSE by **5.4–23.5%**.

Alternatives tested

- ▶ Extra Trees did not replace Ridge after validation.
- ▶ Transition weighting slightly helped transition errors but worsened overall RMSE.
- ▶ A logistic warning classifier had low precision (0.14–0.16) and high false-warning rates (0.26–0.29).

Meaning

The compact hybrid model was retained. The warning classifier remains a research diagnostic, not an operational alarm.

8. A target-free virtual sensor works, with limitations

Selected model

- ▶ 50/50 validation-selected blend
- ▶ Feature Set D Ridge + LightGBM
- ▶ No current or historical target input

Test measure	Mean baseline	Blend
MAE	0.355	0.249
RMSE	0.454	0.301
R^2	≈ 0	0.558

Interpretation

- ▶ RMSE improved by about **33.6%** over the training-mean baseline.
- ▶ The linear and nonlinear models made complementary errors.
- ▶ An empirical 90% interval achieved **89.7%** test coverage.
- ▶ Performance degraded strongly outside the training envelope.

This estimates the current value; it is not yet a validated 10-minute early-warning forecast without target history.

9. The dashboard turns the notebooks into an investigation tool

Available functions

- ▶ Upload a CSV or use the default month
- ▶ Explore signals, missingness and correlations
- ▶ Detect candidate multivariate process-change windows
- ▶ Inspect which sensors drove each window

Prediction functions

- ▶ Target-anchored forecasts at 1, 5, 10 and 15 minutes
- ▶ Target-free virtual-sensor estimates
- ▶ Prediction intervals and drift flags
- ▶ Error metrics only when labels are available

How to read a detected window

The green score measures joint sensor movement, not target abnormality. A highlighted window is a candidate process transition; it must be compared with the later target response, prediction error and plant-event records.

10. Current conclusion and supervisor validation needs

What the evidence supports

- ▶ Target-anchored change forecasting beats persistence at all four horizons.
- ▶ The target-free virtual sensor is materially better than a constant baseline.
- ▶ Process-aware and temporal information add measurable value.

What remains unproven

- ▶ Generalization to another month, season or product grade
- ▶ Reliable operational warning during confirmed plant events
- ▶ Causal or controllable interpretation of important variables

Questions for process validation

Confirm units for acid, ground phosphate, fuel oil, steam, wash liquid and recyclage; define the recycled stream; confirm slurry mass/volume stream correspondence; identify startups, shutdowns and disturbances; confirm which variables are truly controllable.

Recommended next evidence: evaluate a later labeled month against the frozen pipelines and compare stable operation with confirmed transition events.